

Lab Project Report

EE 4170

5-Element Yagi-Uda Antenna Array

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Introduction

The objective of this project was to design and simulate a 5-element Yagi-Uda array optimized for a resonant frequency of 50.1 MHz using CST. The design parameters are derived from Example 10.3 in the Balanis textbook, targeting a directivity of 9.2 dBi. The dimensions for this array were given by this example, so the difficulty of the task stemmed from creating the optimal driving element.

The array consists of one reflector, one driven element, and three directors. Design calculations account for element thickness compensation to ensure accuracy at the operating frequency (50.1 MHz). Listed below are the design formulations and parameters:

At $f_0 = 50.1$ MHz:

- $\lambda = c / f_0 = 5.988\text{m}$
- $d / \lambda = 0.00424\lambda$ (Parasitic element diameter)
- $D / \lambda = 0.00852\lambda$ (Boom diameter)

Element	Final Length (λ)	Spacing (λ)
Reflector (L_1)	0.49	0.2
Director (L_3)	0.447	0.2
Director (L_4)	0.443	0.2
Director (L_5)	0.447	0.2

Procedure

I began with parameterization by defining λ and all element dimensions as variables in CST to allow for easier fine-tuning. This was crucial to creating an effective driven element for this array. To achieve an S11 of -20 dB, modifying the driver was imperative to account for the coupling effects from the parasitic elements.

Next, I had to position the parasitic elements and the driven element in a line. I modeled the elements as cylinders with diameters of 2.54cm. I then applied a discrete port to the center of the driven element and set up a frequency domain solver with a range of 40 - 60 MHz. I used a frequency domain solver for its relative speed in simulation.

I ran an initial simulation to garner a baseline of directivity and input impedance, allowing myself to further optimize the array's S11 value. The S11 was initially terrible, about -10 dB, due to the very low input impedance, likely due to the parasitic elements coupling with the driver. To compensate for this huge decrease in input impedance, I used a folded dipole to multiply that impedance by roughly four times.

Model Geometry

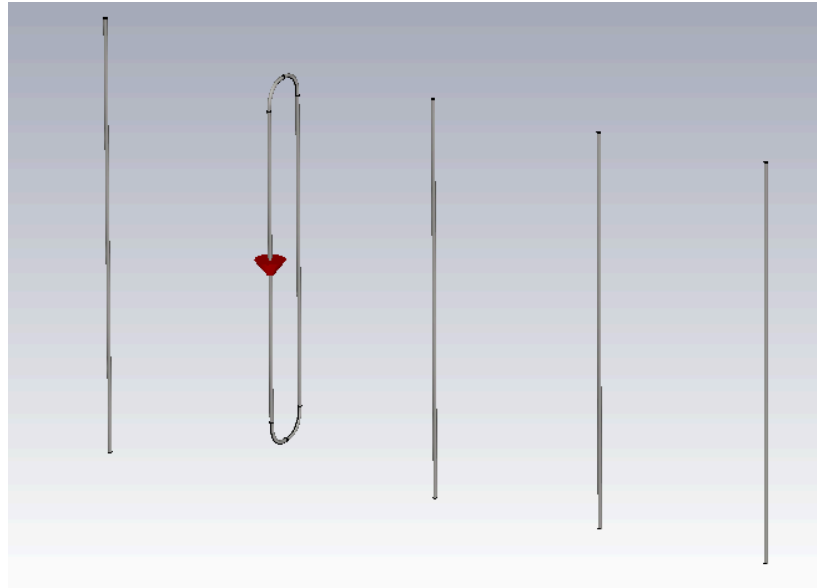


Figure 1: 5-Element Yagi-Uda Array (driven by folded dipole)

3D Schematic 1D Results\Z Matrix\Z1,1			
Parameter List			
Name	Expression	Value	Description
A	$= 0.35 \cdot \lambda$	209.58	
l2	$= 0.45 \cdot \lambda$	269.46	
l1	$= 0.49 \cdot \lambda$	293.412	
l4	$= 0.443 \cdot \lambda$	265.2684	
l3	$= 0.447 \cdot \lambda$	267.6636	
l5	$= 0.447 \cdot \lambda$	267.6636	
diameter	$= 2.54$	2.54	
diameter2	$= 2.54$	2.54	
l2gap	$= 5$	5	
Z0	$= 15$	15	
R	$= 17.6$	17.6	
lam	$= 598.8$	598.8	
<new parameter>			

Figure 2: Final Parameters of Array

Results

The simulation results below provide the input impedance, S11, and radiation characteristics of the designed array. **Figure 3** depicts a 3D model of the farfield pattern of this array, with a directivity of 10.62 dBi, which is notably higher than the value given in Example 10.3 from Balanis.

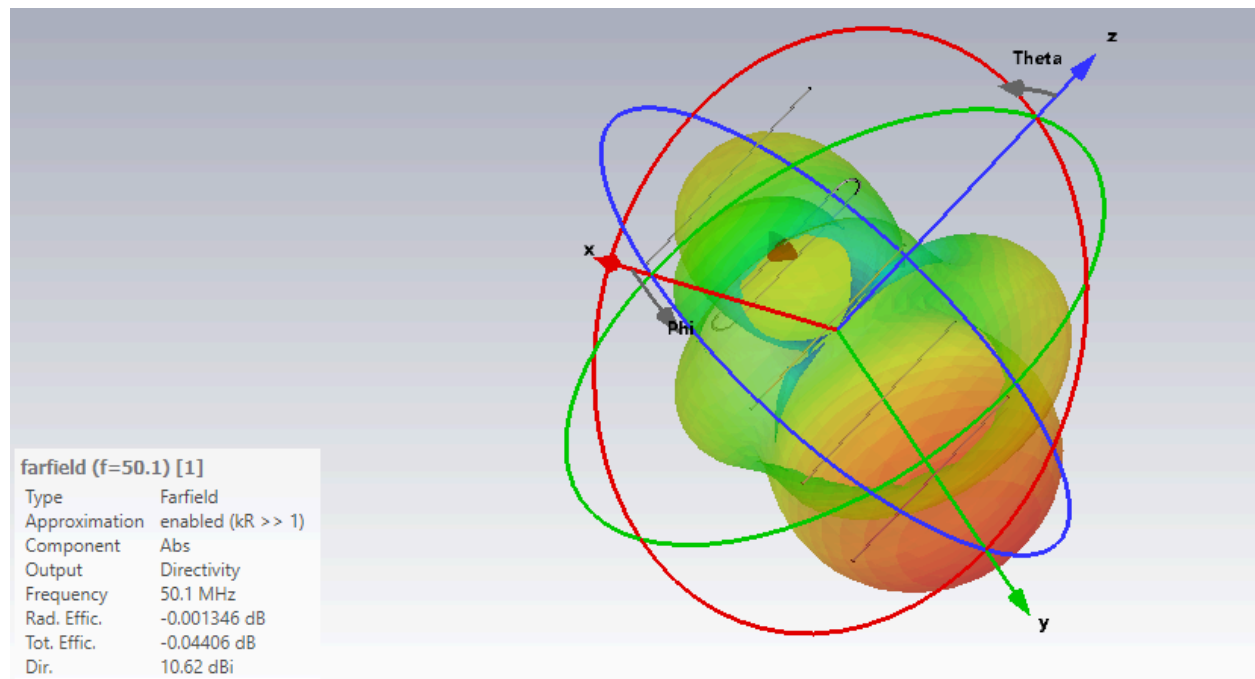


Figure 3: 3D Farfield of 5-element Array

After finding that the directivity of this array was very high, I continued to further optimize the S11 parameter of this array, continuing to move towards achieving the less than -20 dB value I was aiming for. After adjusting the folded dipole just a bit more, I was able to simulate and record an S11 of -20.09 dB (**Figure 4**), as the input impedance was $60.98 + j0.2$ (**Figure 5**) at the resonant frequency. Note that this 60-ohm value is about 4 times larger than the value I ran into when using a standard linear half-wave dipole.

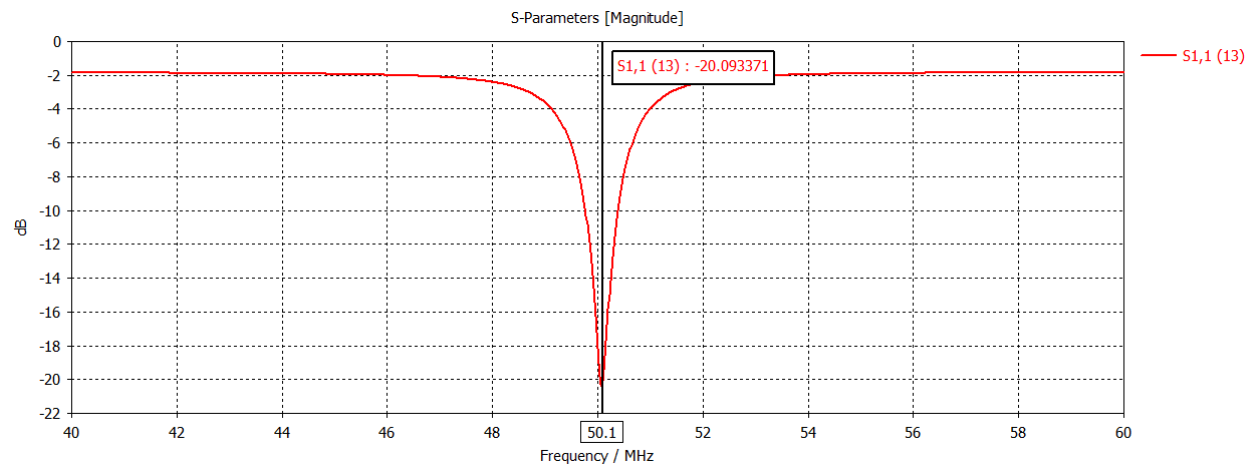


Figure 4: S11 Plot

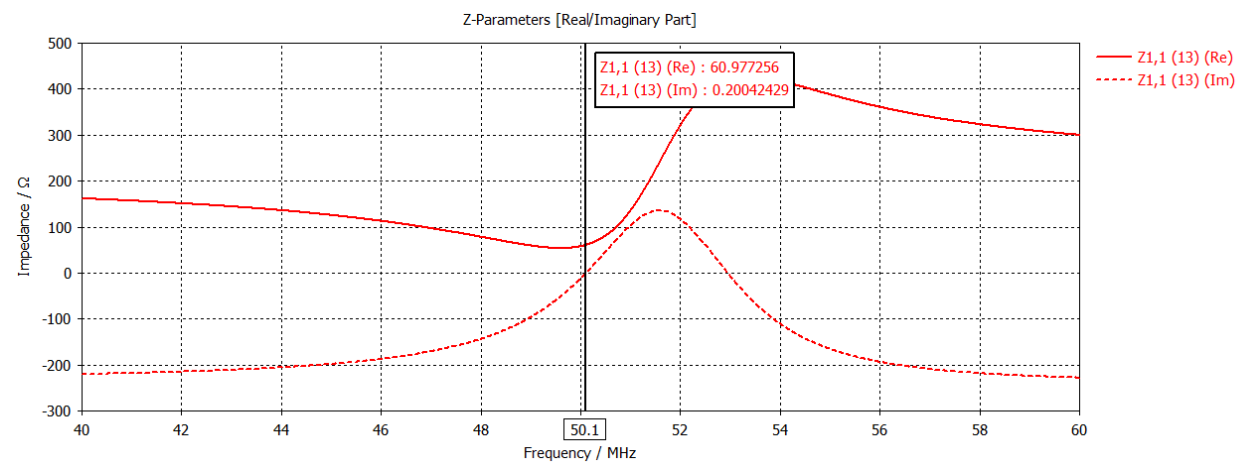
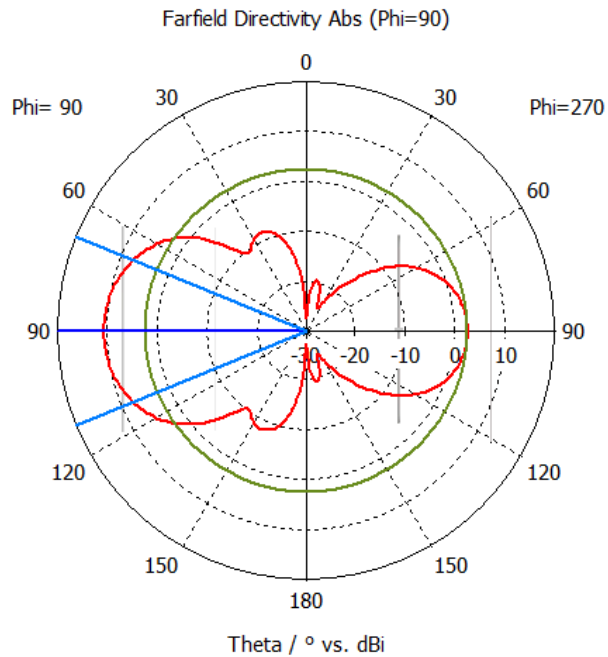


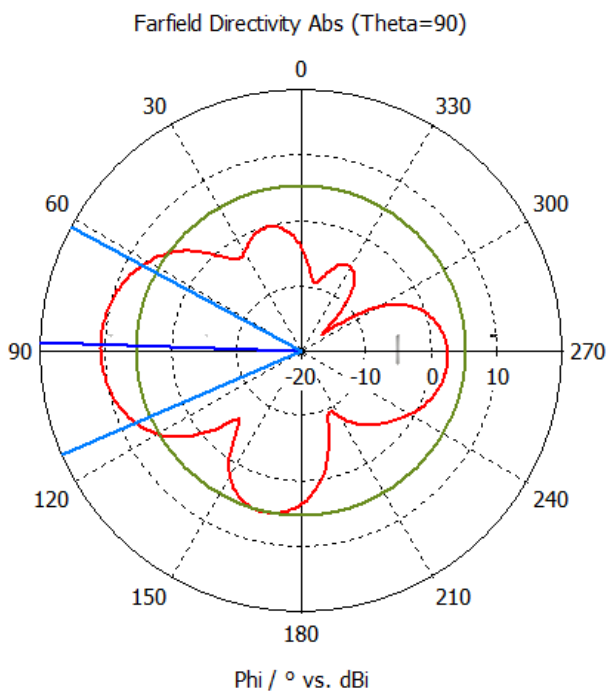
Figure 5: Impedance Plot



— farfield (f=50.1) (13)

Frequency = 50.1 MHz
 Main lobe magnitude = 10.6 dBi
 Main lobe direction = 90.0 deg.
 Angular width (3 dB) = 44.3 deg.
 Side lobe level = -8.2 dB

Figure 6: Farfield E-Plane



— farfield (f=50.1) (13)

Frequency = 50.1 MHz
 Main lobe magnitude = 10.6 dBi
 Main lobe direction = 88.0 deg.
 Angular width (3 dB) = 51.9 deg.
 Side lobe level = -5.3 dB

Figure 7: Farfield H-Plane

The farfield's E- and H-planes are pictured in **Figure 6 and 7** respectively. These show that the end-fire, forward radiation is far more concentrated than other side lobes, indicating that the 5-element array is highly directional and directive in the desired direction, especially compared

to just a simple dipole. It's fascinating how much more directive an array containing the same number of fed elements, with just some parasitic elements, can be.

Conclusion

The simulated 5-element Yagi-Uda array successfully met the design requirements specified in Example 10.3. The directivity achieved in CST was approximately 10.6 dBi, outshining the textbook's stated value of 9.2 dB. The folded dipole, though initially difficult to model, served highly effectively as a feeder, providing a higher input impedance that simplifies matching to standard transmission lines compared to the simple center-fed dipole, leading to a very strong match to the 50-ohm feeding port. The front-to-back ratio and side-lobe levels observed in Figures 6 and 7 confirm the high-gain directional characteristics expected of this specific optimized geometry.